Pentium 4 and Core2 Architecture Notes

1 Processor Evolution

Pentium 4 and Core2 processors are advanced implementations of the Pentium Pro architecture, designed to improve performance through higher clock speeds, improved memory bandwidth, and parallel execution.

1.1 Pentium 4

- Initially released at 1.3 GHz
- Later versions reached speeds up to 3.8 GHz
- Focused on high clock frequency design

1.2 Front-Side Bus (FSB)

- Early versions use 100 MHz base clock
- Quad-pumped bus provides effective speed of 400 MHz
- Later versions use 133 MHz base clock, marketed as 533 MHz

Exam Note: Quad-pumping transfers data four times per clock cycle to increase bus bandwidth.

2 Memory Interface

The memory interface is controlled by the chipset and optimized for high bandwidth using parallel memory channels.

- Uses Intel 945, 965, or 975 chipsets
- Dual-pipe memory bus architecture
- Each pipe is 32 bits wide
- Combined memory width is 64 bits

2.1 Main Memory

- Uses paired DDR2 memory modules
- Supported speeds: 600 MHz, 800 MHz, 1033 MHz

Data Flow:
CPU → L2 Cache → Chipset → Main Memory

3 Modern Interfaces

- Serial ATA (SATA) supported for high-speed disk access
- PCI Express (PCIe) replaces AGP for graphics
- IDE retained for legacy storage devices

4 Register Set Enhancements

4.1 MMX Registers

- Designed for multimedia and signal processing
- Separate from traditional floating-point registers

4.2 XMM Registers (SIMD)

- Eight registers: XMM0–XMM7
- Each register is 128 bits wide
- Supports SIMD operations
- Can store:
  - Two 64-bit double-precision values, or
  - Four 32-bit single-precision values

5 Hyper-Threading Technology

Hyper-Threading allows a single physical processor to appear as two logical processors.

- Two execution units, each with its own register set
- Shared Bus Interface Unit (BIU)
- BIU contains cache and memory interfaces
- Improves CPU utilization during stalls

Exam Note: Hyper-Threading does not create two physical processors.

6 Multiple-Core Technology

Multiple-core processors integrate more than one independent processor core on a single chip.

6.1 Dual-Core

- Pentium D: separate caches per core
- Core2 Duo: shared cache (2 MB or 4 MB)

6.2 Quad-Core

- Four independent cores
- Enables true parallel execution

- Clock frequency stabilized around 3–4 GHz
- Performance improvement achieved via parallelism

7 Model-Specific Registers (MSRs)

- Total MSRs: 1743
- Address range: 000H to 6CFH
- Used for hardware control and system management

7.1 Access Mechanism

- ECX holds MSR number
- EDX:EAX used for 64-bit data transfer

8 Performance-Monitoring Registers (PMRs)

- Used to measure processor performance
- Count cycles, instructions, and events
- Time-Stamp Counter (TSC) accessible to user programs
- PMRs are read-only

9 64-Bit Extension Technology

Intel extends the x86 architecture to 64-bit while maintaining backward compatibility.

9.1 General-Purpose Registers

- Total registers: 16
- Legacy registers extended to 64 bits (RAX–RDI)
- New registers added: R8–R15

9.2 Register Access

- Registers accessible as 64, 32, 16, or 8 bits
- No high-byte registers for new registers

10 Segmentation in 64-Bit Mode

- Segmentation is mostly disabled
- CS base address set to zero
- DS, ES, SS base addresses ignored
- Flat memory model is used

11 Paging in 64-Bit Mode

- 64-bit linear address translated to 52-bit physical address
- Supports up to 4 PB physical memory
- Virtual address space up to 16 EB
- Uses four-level page table hierarchy

Exam Summary: In 64-bit mode, segmentation is minimized and paging becomes the primary memory management mechanism.